

International Islamic University Chittagong
Center For General Education (CGED)
Midterm Examination Spring-2025
Course Code: GEBL-2401
Course Title: Bengali Language and Culture
Full Marks:30 Time:1.5 Hours

ক-বিভাগ
ভাষা ও নির্মিতি : ২০
(প্রতিটি প্রশ্নের মান সমান)

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০১.	“বাংলা ভাষার উদ্ভব ও বিকাশ” - শীর্ষক একটি নিবন্ধ উপস্থাপন কর ।	১০	CLO1	Create
০২.	নিচের পরিভাষাগুলোর পরিচয় দাও: ক) ধ্বনি খ) বাংলা মৌলিক স্বরধ্বনি গ) স্পর্শ ব্যঞ্জন ঘ) মহাপ্রাণ ধ্বনি ঙ) যৌগিক স্বরধ্বনি ।	৫x২=১০	CLO1	Understand
অথবা				
	“নারীর প্রতি সহিংসতারোধ” - শিরোনামে সংবাদপত্রে প্রকাশের উপযোগী একটি প্রতিবেদন উপস্থাপন কর ।	১০	CLO1	Create

খ-বিভাগ
সংস্কৃতি: ১০

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০১.	“জীবনে এমন কত বিচ্ছেদ, কত মৃত্যু আছে, ফিরিয়া ফল কী।” - উক্তির আলোকে রবীন্দ্রনাথ ঠাকুরের পোস্টমাস্টার গল্পের কেন্দ্রীয় চরিত্র বিশ্লেষণ কর ।	১০	CLO2 CLO3	Analyze

International Islamic University Chittagong
Department of Computer Science and Engineering

B. Sc. in CSE Mid-term Examination, Spring 2025

Course Code: CSE 2423

Course Title: Database Management System

Total marks: 30

Time: 1 hour 30 minutes

				CL	D
				O	L
Q1	a)	Differentiate between the File System and Database System with respect to data redundancy, concurrency, and data integrity.	3	1	2
	b)	Discuss the roles of the Storage Manager and how it interacts with other components of a DBMS.	4	2	2
	OR	Explain Query processing system of DBMS with appropriate figure			2
	c)	List and explain any four types of database systems. Highlight the key features and differences among them and indicate their application areas	3	1	2
Q2	a)	Design an ER Diagram for a university system that manages Students, Courses, and Instructors. A student can enroll in multiple courses, and each course can be taught by multiple instructors. Use appropriate entities, relationships, attributes, cardinalities, and indicate any weak entities.	5	3	3
	b)	Analyze the impact of generalization and specialization when modeling faculty as either "Full-Time" or "Part-Time".	3	2	4
	c)	What is a composite key? Give an example from your ERD.	2	2	2
Q3	a)	What is the difference between relational algebra and SQL?	6	2	3

Given Schema:

Student(SID, Name, DeptID)

Course(CID, Title, Credits)

Enrolled(SID, CID, Grade)

Write relational algebra expressions for the following:

1. Get names of students enrolled in the course titled "DBMS".
2. Find students who have not enrolled in any course.

OR What is the difference between relational algebra and SQL? Using above schema, write relational algebra expressions for the following:

1. Get the names of students who have received a grade of 'A'
 2. Find names of students enrolled in at least one course from DeptID = 'CSE'.
- b) Use relational algebra expressions along with appropriate example tables to illustrate how these operations can be combined to retrieve meaningful information from a database. 4 2 3

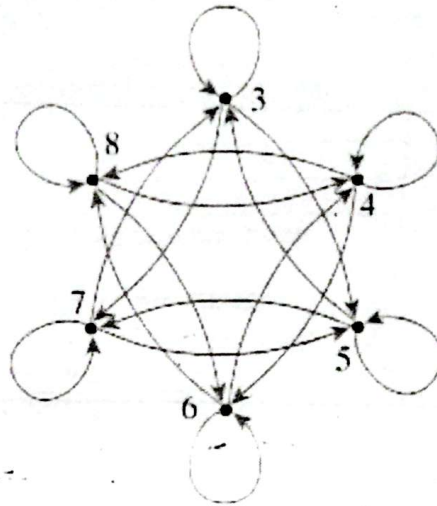
International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
B. Sc. in CSE, Mid Term Examination, Spring-2025
Course Code: MATH-2407, Course Title: Mathematics-IV

Time: 1:30 Hours

Marks: 30

[Answer the following questions. Figures in the right margin indicates full marks]

- | | | | |
|--|------------|-------------|----------|
| 1. a) Let the function $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be defined by $y = f(x) = x^2 + x - 1$ then find the value of $f^{-1}(10)$ | Marks
2 | CLO
CLO1 | DL
C2 |
| b) | 4 | CLO1 | C2 |



Determine whether the above relation is Anti-symmetric and Transitive

- | | | | |
|---|---|------|----|
| c) Using Demoivre's theorem find the quadratic equation whose roots are the n-th power of the roots of the equation, $x^2 - 2x \cos \theta + 1 = 0$ | 4 | CLO3 | C2 |
|---|---|------|----|

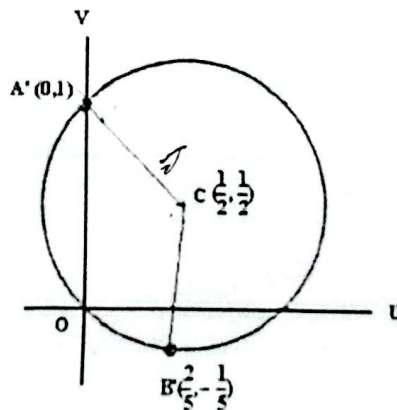
Or

If $(1 + i\frac{x}{a})(1 + i\frac{x}{b})(1 + i\frac{x}{c}) \dots = A + iB$, Then prove that

$$(1 + \frac{x^2}{a^2})(1 + \frac{x^2}{b^2})(1 + \frac{x^2}{c^2}) \dots = A^2 + B^2$$

4 CLO3 C2

- | | | | |
|---|---|------|----|
| 2. a) Map the straight line joining $A(-2 + j)$ and $B(3 + j6)$ in the z-plane on to the w-plane when $w = u + jv = f(z) = 3 + j2z$ | 6 | CLO1 | C2 |
| b) Prove that $f(z) = z ^2$ is not harmonic function. | 2 | CLO1 | C3 |
| c) | 2 | CLO1 | C2 |



Write the equation of the graph above.

3. a) Evaluate $\int_C z dz$ where C is given by $x = 3t, y = t^2$ with $-1 \leq t \leq 4$

4 CLO1 C3

b) Using Cauchy's Integral Formula evaluate $\int_C \frac{z^2 - z + 1}{z - 1} dz$ Where c is the circle for $|z| = 1$

4 CLO1 C3

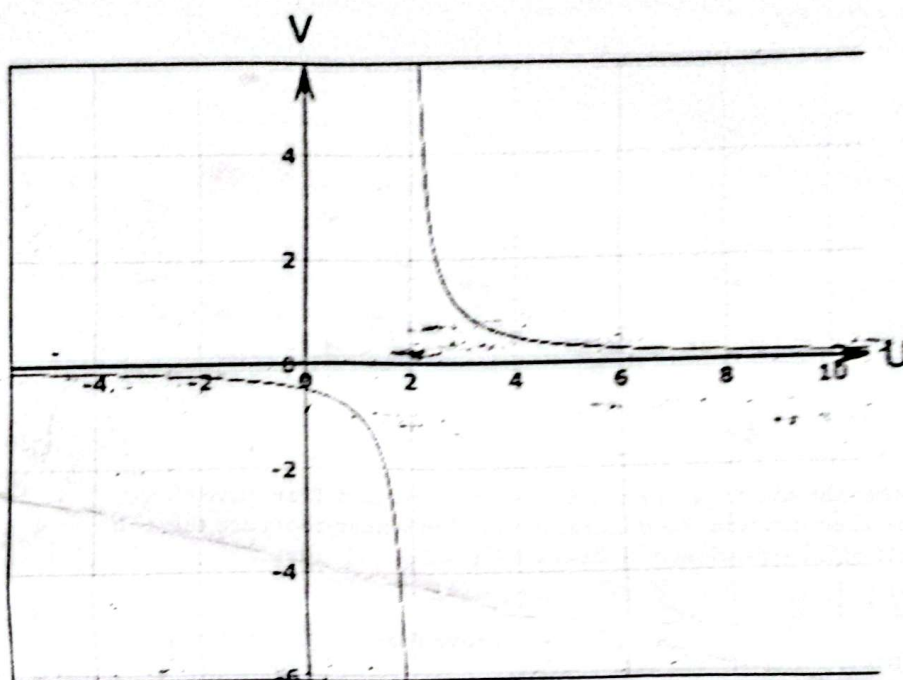
Or

Evaluate $\int_C \frac{2z + 1}{z^2 + z} dz$ Where c is the circle $|z| = \frac{1}{2}$

4 CLO1 C3

2 CLO1 C2

c)

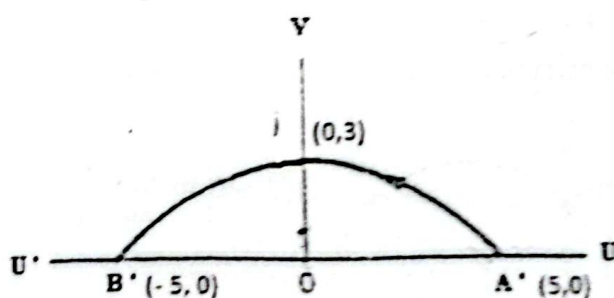


Where singularity occurs?

Or



2 CLO1 C2



Write the equation of the graph above



International Islamic University Chittagong Department of Computer Science and Engineering <i>B. Sc. in CSE Midterm Examination, Spring 2025</i> Course Code: CSE 2421 Course Title: Computer Algorithms Total marks: 30 Time: 90 minutes			
[Answer all the questions; in some questions, there might be options; Figures in the right hand margin indicate full marks.]			
	CO	DL	
1.			
a) Find the complexity of the following two functions. <pre>IFun(int n) for (j = 1; j < n; j *= 2) { int k = 0; while (k <= sqrt(n)) k++; } RFun(int b, int p) { if (p == 0) return 1; else return b * RFun (b, p - 1); }</pre>	CO2	N	3
b) Find time complexity by applying recursive tree method: $T(n) = 3T(n/4) + cn^2$ OR Find time complexity by applying substitution method: $T(n) = 2T(n/2) + n$	CO2	N	3
c) Assume that $f(n) = 4n + 10$ and $g(n) = 3n^2 + 5$. Is $g(n)$ an upper bound of $f(n)$? Prove or disprove this.	CO2	N	2
d) Find the asymptotic bound of the following recurrence by using master method: $T(n) = 4T(n/2) + n^2$	CO2	N	2
2.			
a) Assume you have a max-heap with n elements stored in an array $A[1 \dots n]$. You are told that $A[10] = 500$. What is the maximum number of elements in the heap that could be greater than 500? Justify your answer.	CO1	N	2



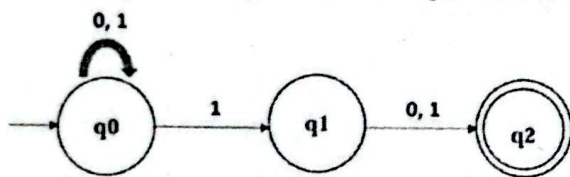
b)	When does Quicksort perform much slower than Merge-Sort? Why does this happen? Illustrate with an example.	CO1	E	3
c)	Consider the following array of non-negative integers: $P = [4, 2, 2, 8, 3, 3, 1, 5]$. Apply the Counting Sort algorithm to sort this array. Show intermediate steps.	CO1	U	2
d)	You are given the following array: $C = [12, 4, 9, 5, 11, 21, 2, 10]$. Perform the first partition operation of the quicksort algorithm using the last element as the pivot. Clearly show: (i) The initialization of the pointers (ii) Each comparison and any swaps that occur during the partitioning process (iii) The final arrangement of the array after placing the pivot in its correct position. (iv) Indicate the index of the pivot after the partitioning step. OR You are given the following array: $E = [35, 25, 40, 4, 10, 34, 11, 13]$. Perform the Heap Sort Algorithm on the array. i) Provide the state of the array after each major step, especially after building the max-heap and after each extraction + re-heapification. ii) Indicate the final sorted array after completing the heap sort process.	CO1	U	3
3.				
a)	Find an optimal parenthesization of a matrix chain product whose sequence of dimension is $\{4, 8, 5, 6, 3\}$. Show calculations of each steps. OR Find the longest common subsequence of the following two sequences using dynamic programming and show the steps. $X = \{ACGTAB\}$ $Y = \{GATXAYB\}$	CO4	U	4
b)	What is overlapping subproblem? Do you think that the problem of calculating Fibonacci number shows this property? Justify your answer.	CO4	N	3
c)	Discuss the key differences between Divide & Conquer and Dynamic Programming. In what scenarios would each of these algorithms be preferable, and why? Provide examples to support your answer.	CO4	E	3

Ass - 21 cr
 wa - 31 cr
 number - 100
 rev - 100
 Exp - 100

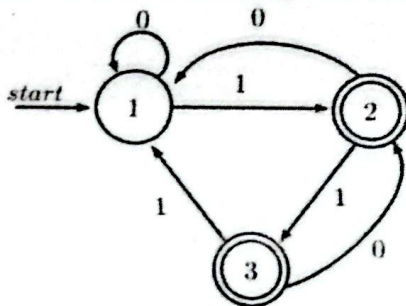
[Answer all the questions; in some questions, there might be options;
 Figures in the right hand margin indicate full marks.]

1.
 - a) What is the subject matter of Computability Theory? 1
 OR
 What is the subject matter of Complexity Theory?
 - b) Construct DFA for the following languages where alphabet is $\{0, 1\}$. 4.5
 - i. $\{w \mid w \text{ length of string is not divisible by } 2\}$
 - ii. $\{w \mid w \text{ does not contain the substring } 0110\}$
 - iii. $\{w \mid w \text{ must not ends with } 011\}$
 - c) Write regular expressions for the languages described in 1(b). 4.5

2.
 - a) Convert the following NFA to an equivalent DFA. 5



- b) Convert the following DFA to a regular expression: 5



3.
 - a) What are the components of a DFA? What is the difference between the transition function of NFA and DFA? 4
 OR
 Formally define a regular expression.

- b) Define *string* and *language*. 1
 - c) What are the languages described by the following languages? Write a one sentence description for each language. 3
 - i. $(00 \cup 1)^*$
 - ii. $(0 \mid 1)^*111$
 - iii. $1^*(01)^*$

OR

Construct NFA for the following languages where alphabet is $\{0, 1\}$.

- i. $\{w \mid w \text{ contains the substring } 0101\}$
- ii. $\{w \mid w \text{ begins with a } 1 \text{ and ends with a } 0\}$

- d) Convert the following regular expressions to nondeterministic finite automata. 2

- i. $(00 \cup 1)^*(11)^*$
- ii. $(1^* \cup 0(00)^*)111$

OR

Construct NFA that recognizes the concatenation of the languages described in 1(b)(i) and 1(b)(ii).

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. Engineering in CSE
Midterm Examination, Spring 2025

Course Code: ACC 2401

Course Title: Financial Managerial Accounting

Time: 1 hour 30 minutes

Full Marks: 30

Answer all the questions. The figures in the right-hand margin indicate full marks.

Course Outcomes (COs), Program Outcomes (POs) and Bloom's Levels (BL) of the Questions			
CO	CO Statements	PO	BL
CO1	Explain the basic concept of financial accounting, cost accounting and management accounting.	PO11	C2
CO2	Analyze the basic concept of Cost Accounting and preparation of Cost Sheet.	PO11	C4
CO3	Apply the tools from accounting and cost accounting this would facilitate the decision making i.e. Budgeting, Make or Buy decision.	PO11	C3
CO4	Compare the different business situations and suggest to best solution with analytical abilities.	PO11	C5

Bloom's Levels (BL) of the Questions						
Letter Symbols	C1	C2	C3	C4	C5	C6
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	"An internal claim in a business is owners' equity." Put your detailed accounting equation explanation here along with your explanation of this statement.	CO1	C2	6
1)	b)	Discuss whether the following events are transaction or not with their reasons: I. A business signs a contract with a supplier for 40000 tk. II. Owners pays 34000 tk of personal funds for a vacation of her family III. Purchased stationary items of 2000 tk on account IV. Business decides to hire 2 executives that will cost 100000 tk per month	CO1	C2	4
2)	a)	Lisa Reed owns and operates an interior design studio called Reed Interiors. The following amounts summarize the financial position of her business on August 31, 19X2: During September 19X2 the following events occurred: a. Reed inherited \$15,000 and deposited the cash in the personal bank account. b. Performed services for a client. c. Paid off the beginning balance of accounts payable. d. Purchased supplies on account, \$500. e. Collected cash from a customer on account, \$1,000. f. Invested personal cash of \$1,000 in the business. g. Consulted on the interior design of a major office building and billed the client for services rendered, \$2,400. h. Paid office rent - \$900 and Paid advertising - \$100. i. Sold supplies to another business for \$150 cash, this was the cost of the supplies. j. Withdrew cash of \$(Last Four digit of your student ID) for personal use. k. Collected cash from his friend for buying a personal car \$1,00,000. l. Reed deposited \$15,000 cash to his personal bank account by selling some common stock.	CO2	C4	10

AGE 21 D J C
R.L. Wi. 21. C 20 D

		Required 1. Analyze the effects of the above transactions on the accounting equation of Reed Interiors 2. Prepare the income statement and statement of owner's equity of the Reed Interiors' business based on the above transaction..			
3)	a)	Gill brothers, the law research firm, opened a business in the name of Gills Brothers in June 1, 2024. Here are the following events that occurred during the month of June: June 1: Received cash investments from owner 2,00,000 tk June 4: Purchased some equipment for 40000tk on account. June 15: Performed some legal services and received cash 8000 tk June 17: Owners received 20000 tk from her former business named apparel company June 20: Paid expense: office rent 4000 tk, insurance premium 1000 tk, utility 2000 tk June 26: Sold equipment for 30000 tk which was the cost of the equipment June 28: Owners withdrew 3000 tk from business for paying his children's school fees Required: 1) Record each transaction in the journal. 2) Post the transaction to the ledger with label the balance of each account..	CO2	C5	10
OR					
3)	a)	Andrea Scarlett is a realtor. She organized her business as a corporation, Andrea Scarlett, Realtor, P.C. (Professional Corporation) 1. The business owes \$61,000 on a note payable for land that the business acquired for a total price of \$83,000. 2. The business spent \$23,000 for a Zinka Banker real estate franchise, which entitles the business to represent itself as a Zinka Banker office. This franchise is a business asset. 3. Scarlett owes \$80,000 on a personal mortgage for her personal residence, which she acquired in 2012 for a total price of \$160,000. 4. Scarlett has \$5,000 in her personal bank account, and the business has \$9,000 in its bank account. 5. Scarlett owes \$(Last Four digit of your student ID) on a personal charge account with Chico's. 6. The office acquired business furniture for \$15,000 on Sep 25. Of this amount, the business owes \$2,000 on account at September 30. 7. Office supplies on hand at the real estate office total \$1,300. 8. Received partial payment from client on account, \$2700 9. Received \$1,500 cash for helping a client. 10. Scarlett received a birthday gift from his friend valued \$3000. Required 1. Post the transaction to the ledger with label the balance of each account. 2. Prepare a Trial Balance .	CO2	C5	10